

Sparse, Low-Rank, and Deep

A Unified Compressed-Sensing Framework for Music Source Separation
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Problem & Contribution

Music source separation recovers stems (vocals, accompaniment) from a single mixture. Classical decompositions and large neural networks are usually treated as separate worlds. **We show they are points on one axis: recovery of structured (sparse / low-rank) spectrogram representations.**

- ▶ Four separators on a shared STFT front end: low-rank+sparse, supervised NMF, discriminative K-SVD, and a **deep unrolled sparse coder**.
- ▶ The deep sparse coder is the **strongest non-oracle separator and the fastest learned model**.
- ▶ Honest **coherence analysis**: where the recovery theory binds and where it goes slack.

The structured-recovery view

On the magnitude STFT $\mathbf{M} = |\text{STFT}(y)|$: vocals are **sparse** on a formant scaffold; accompaniment is **low-rank** (sustained harmony).

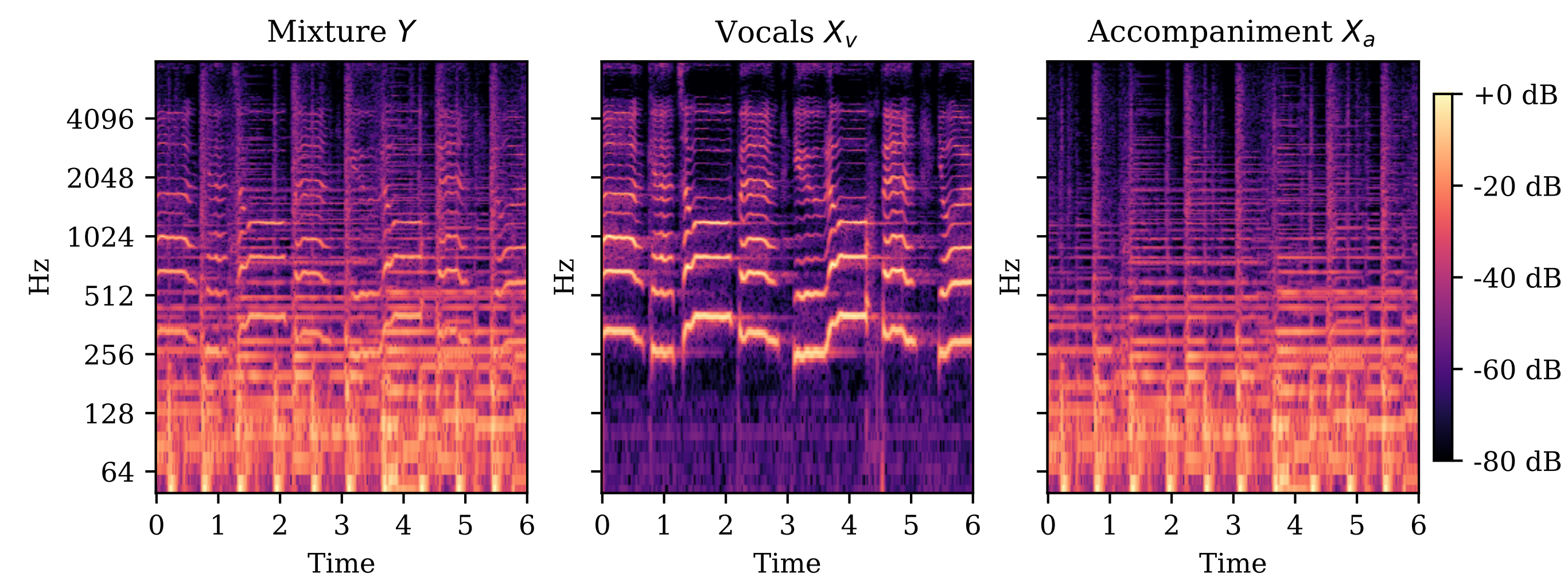


Figure 1: Mixture vs. vocals vs. accompaniment. The structure each method exploits.

Preliminaries: when does recovery hold?

For $\mathbf{y} = \mathbf{A}\mathbf{x}$, $\mathbf{x}$ s -sparse, the convex program $\min \|\mathbf{z}\|_1$ s.t. $\mathbf{A}\mathbf{z} = \mathbf{y}$ recovers $\mathbf{x}$ under any of:

$$\text{spark}(\mathbf{A}) > 2s, \quad \delta_{2s} < \sqrt{2} - 1, \quad s < \frac{1}{2} \left(1 + \frac{1}{\mu(\mathbf{A})} \right).$$

RIP: $(1 - \delta_s) \|\mathbf{x}\|_2^2 \leq \|\mathbf{A}\mathbf{x}\|_2^2 \leq (1 + \delta_s) \|\mathbf{x}\|_2^2$. Coherence $\mu(\mathbf{A}) = \max_{i \neq j} |\langle \mathbf{a}_i, \mathbf{a}_j \rangle|$. Random $\mathbf{A}$ needs $m \geq Cs \log(N/s)$ measurements.

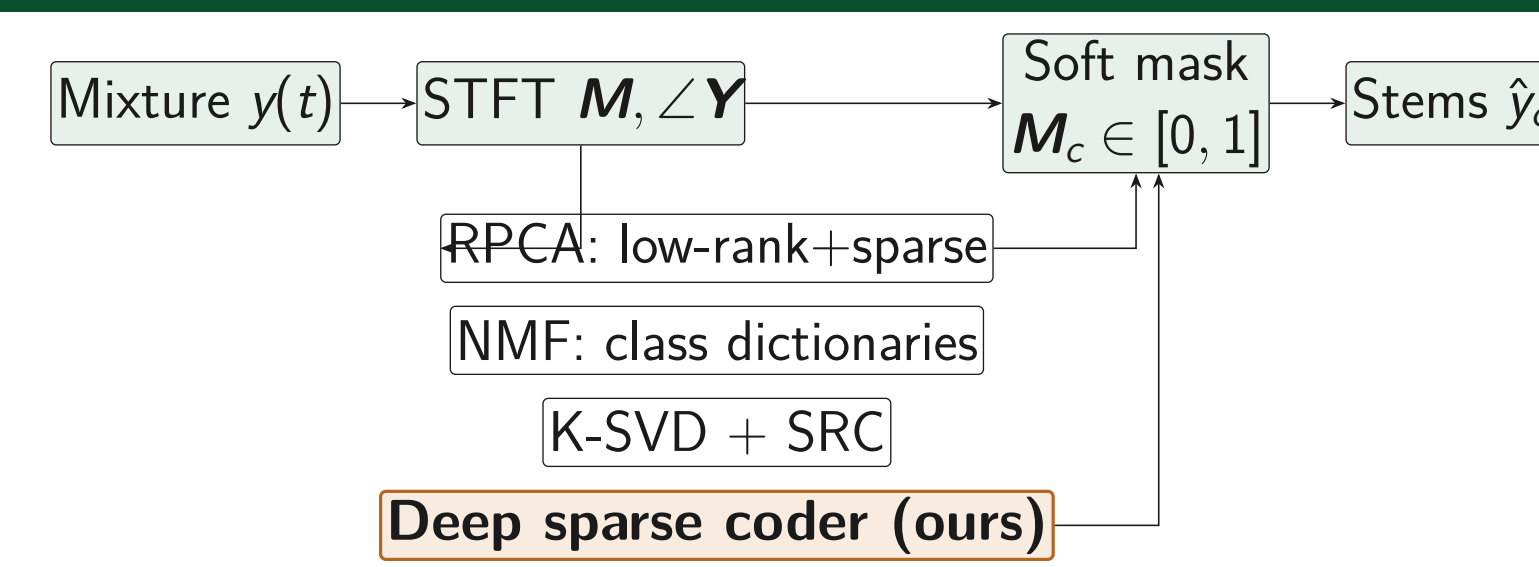
Low-rank: nuclear norm $\|\mathbf{X}\|_* = \sum_i \sigma_i$ is the convex surrogate for rank; principal component pursuit splits $\mathbf{M} = \mathbf{X} + \mathbf{E}$.

Joint sparsity for stereo (MMV)

Stereo channels share active sources, so their codes share a **row support**. Stack per-frame channel spectra into $\mathbf{Y} = [\mathbf{y}_L \ \mathbf{y}_R]$ and run **simultaneous OMP**, scoring atoms by the row norm $\|(\mathbf{D}^\top \mathbf{R})_{n,:}\|_2$ so both channels agree on active atoms. Mixed-norm surrogate $\|\mathbf{X}\|_{1,2} = \sum_n \|\mathbf{X}_{n,:}\|_2$. Shared support *improves* identifiability:

$$s < \frac{1}{2} (\text{spark}(\mathbf{A}) - 1 + \text{rank}(\mathbf{Y})).$$

Unified pipeline



Four interchangeable separators share one front end; the deep coder predicts the mask directly.

Classical recovery stages

Low-rank + sparse (RPCA). Principal component pursuit,

$$\min_{\mathbf{X}, \mathbf{E}} \|\mathbf{X}\|_* + \lambda \|\mathbf{E}\|_1 \text{ s.t. } \mathbf{X} + \mathbf{E} = \mathbf{M}, \quad \lambda = \frac{1}{\sqrt{\max(F, T)}},$$

solved by inexact ALM (SVT + soft-threshold). Low-rank $\mathbf{X}$ =accompaniment.

Supervised NMF. Per-source non-negative dictionaries, KL multiplicative updates $\mathbf{H} \leftarrow \mathbf{H} \odot \frac{\mathbf{D}^\top (\mathbf{M} \oslash \mathbf{D} \mathbf{H})}{\mathbf{D}^\top \mathbf{1}}$; fix $\mathbf{D} = [\mathbf{D}_v | \mathbf{D}_a]$ at test time.

K-SVD + SRC. Overcomplete dictionaries via rank-one atom updates; route each frame by $\mathbf{c}^* = \arg \min_{\mathbf{c}} \|\mathbf{y} - \mathbf{D}_c \mathbf{c}\|_2$.

Deep sparse coding network (novel)

Predict the mask by **unrolling a non-negative proximal solver and learning its dictionaries**. Each composite module: fat (upsampling) then tall (downsampling) sparse code,

$$\boldsymbol{\alpha}^{(1)} = \arg \min_{\boldsymbol{\alpha} \geq 0} \frac{1}{2} \|\mathbf{x} - \mathbf{D}^{(1)} \boldsymbol{\alpha}\|_2^2 + \lambda_1 \|\boldsymbol{\alpha}\|_1 + \frac{\lambda_2}{2} \|\boldsymbol{\alpha}\|_2^2,$$

$$\boldsymbol{\alpha}_k = \left[\mathbf{z}_k - \frac{1}{L} \mathbf{D}^\top (\mathbf{D} \mathbf{z}_k - \mathbf{x}) - \frac{\lambda_1}{L} \right]_+ \quad (\text{unrolled FISTA}).$$

Stack 3 modules + sigmoid head; train end-to-end on

$\|(\mathbf{M}_v - \mathbf{M}_v^{\text{IRM}}) \odot \mathbf{M}\|_1 + \beta \sum_{\ell} \|\boldsymbol{\alpha}^{(\ell)}\|_1$. This is LISTA specialized to non-negative codes and a masking output.

Learned dictionaries

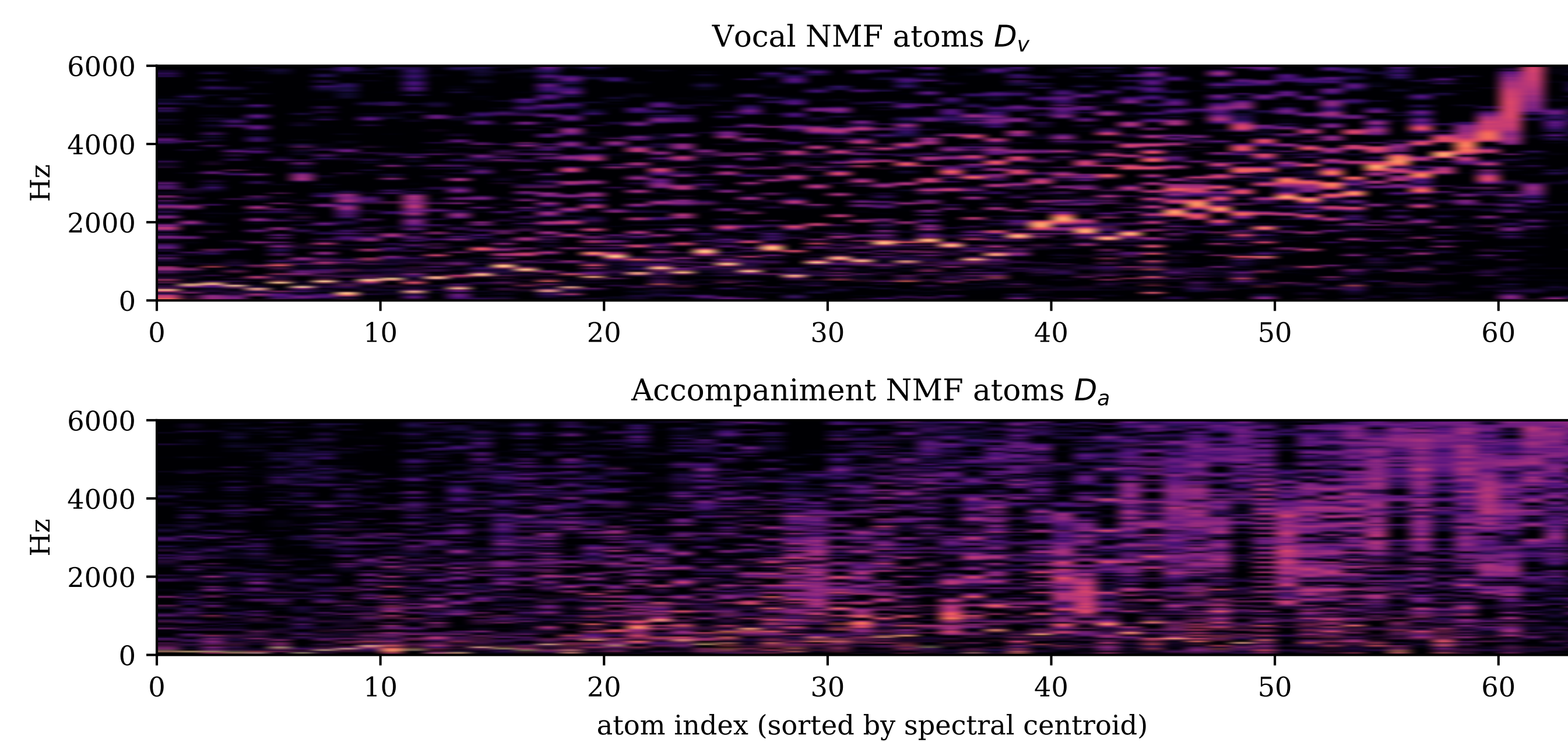
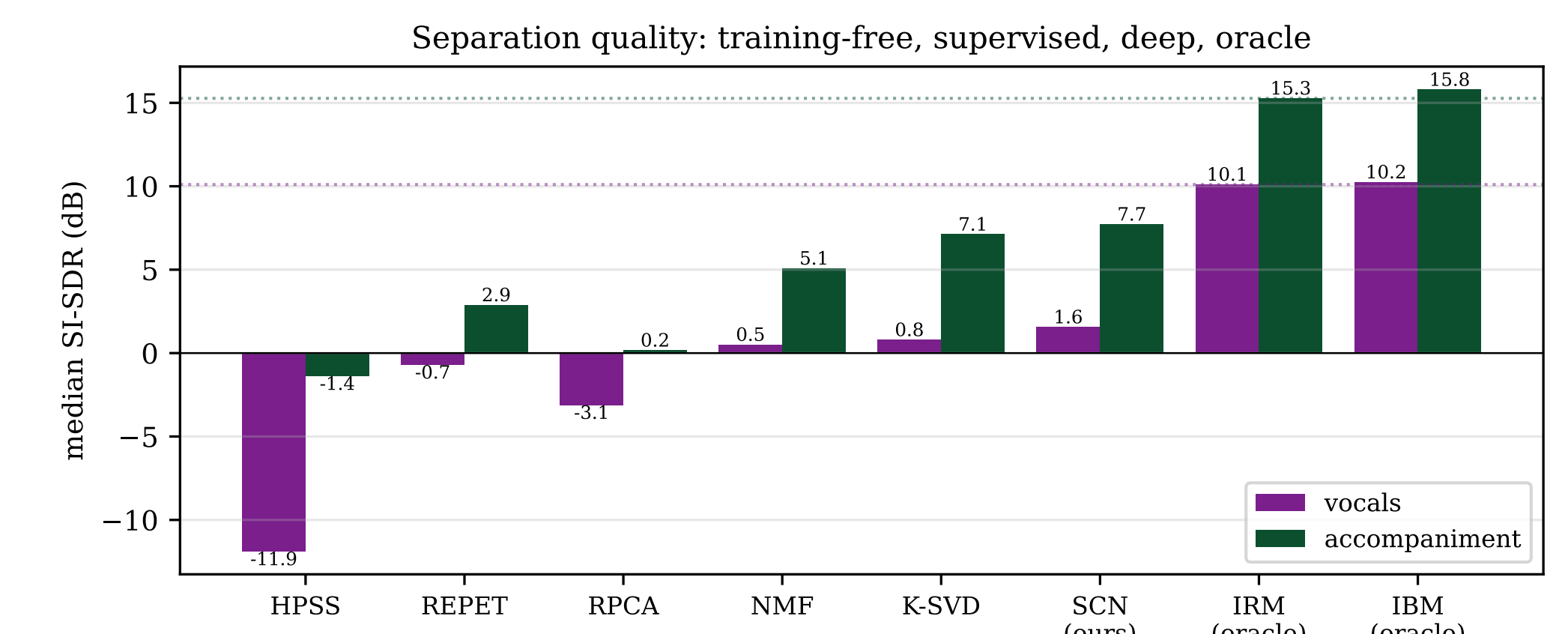


Figure 2: NMF atoms sorted by centroid: vocal (top) ride the formant region, accompaniment (bottom) spread lower. Parts-based, as predicted.

Main results (25 MUSDB18 test tracks)

Method	SI-SDR (dB)		Time (s)
	Vocals	Accomp.	
HPSS (median)	−11.9	−1.4	0.25
REPET-SIM	−0.7	2.9	0.08
RPCA	−3.1	0.2	1.14
NMF (supervised)	0.5	5.1	0.46
K-SVD + SRC	0.8	7.1	0.28
Deep coder (ours)	1.6	7.7	0.08
Oracle IRM / IBM	10.2	15.8	—



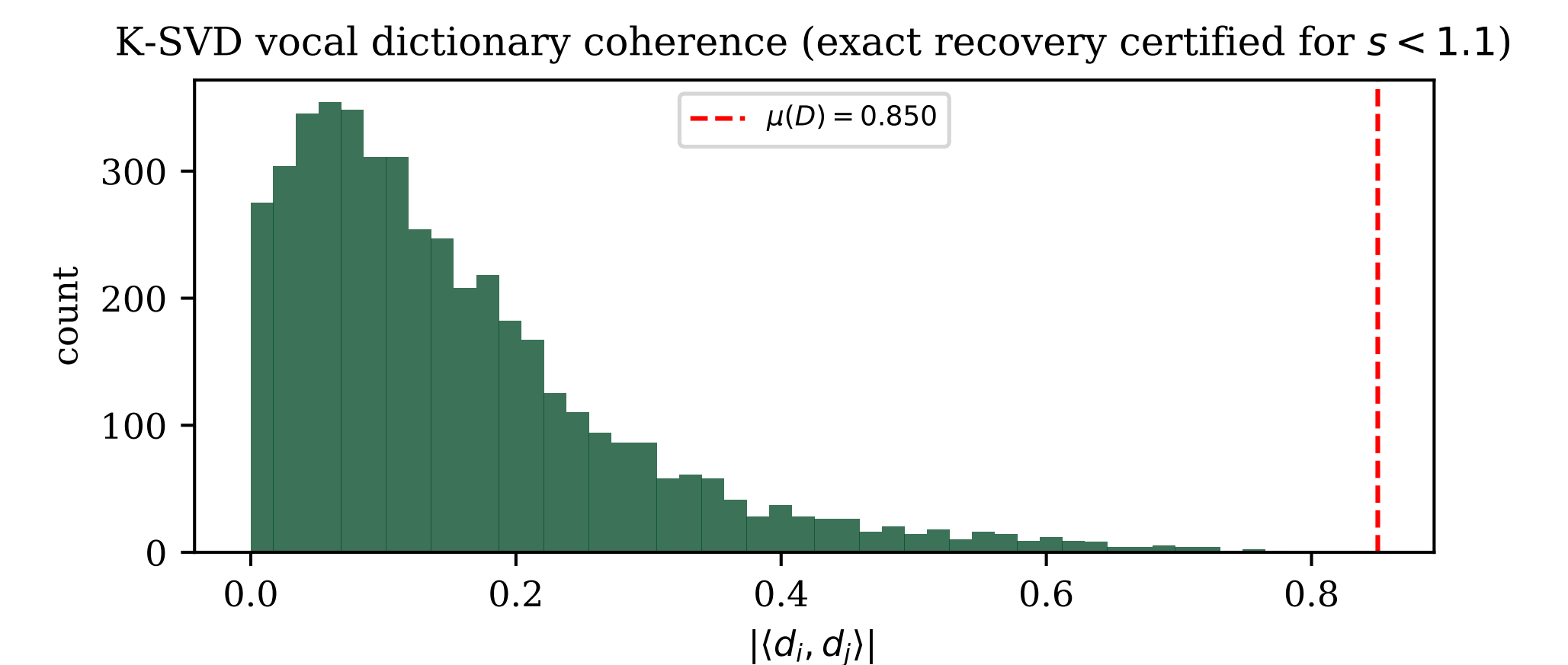
Training-free < supervised < deep. The deep coder is the best non-oracle on both sources and the fastest learned model.

Ablations

NMF atoms K	16	32	64	128
Vocal SI-SDR	2.2	1.9	1.8	1.5
K-SVD s	3	5	10	15
Vocal SI-SDR	1.4	2.3	2.7	2.7
RPCA λ scale	0.5	1.0	1.5	2.0
Vocal SI-SDR	−3.9	−3.6	−3.2	−2.3

Smaller NMF dictionaries *generalize better* (regularization); K-SVD improves with sparsity to $s=10$; larger RPCA λ yields cleaner vocals.

Coherence: an honest negative result



Learned dictionaries are coherent ($\mu \approx 0.85$), so the worst-case bound certifies exact recovery only for $s \lesssim 1$, yet OMP at $s=10$ separates well. **Average-case $\neq$ worst-case** for trained dictionaries.

Takeaways

- ▶ One STFT front end unifies low-rank, NMF, dictionary, and deep methods.
- ▶ **Unrolled deep sparse coding**: best non-oracle, fastest learned model.
- ▶ Increasing structure-learning $\Rightarrow$ monotone quality gain.
- ▶ Code + models + audio:
github.com/takakhoo/ENG5109_Final_Project